

MP352 – Special Relativity

Problem Set 2

Due by the end of lecture on Friday, 13 March 2015

(Throughout this assignment, assume all frames are inertial and all velocities are constant.)

1. The classic 1988 anime video series *Gunbuster* (original Japanese: *Toppu o Nerae!*) is notable not just for being excellent (in my opinion), but also for dealing with the effects of relativity very realistically. For example, in the final episode, protagonists Noriko and Kazumi have just saved the Earth by blowing up some alien bad guys with their giant robots and start heading home. However, they quickly realise that even though the journey will only take two days for them, it'll be 12,000 years later on Earth when they arrive.
 - (a) How close to the speed of light are Noriko and Kazumi travelling? In other words, if their speed is v , determine $1 - v/c$.
 - (b) According to someone on Earth, how far (in light-years) did they travel? And how far was it according to Noriko and Kazumi?
2. Suppose the frame $\mathcal{S}'$ moves relative to frame $\mathcal{S}$ with speed v in the x -direction, and frame $\mathcal{S}''$ moves relative to $\mathcal{S}'$ with speed u' in the x' -direction. The speed u which $\mathcal{S}''$ moves at relative to $\mathcal{S}$ is

$$u = \frac{u' + v}{1 + \frac{u'v}{c^2}}.$$

Show that this gives

$$\gamma(u) = \gamma(u') \gamma(v) \left(1 + \frac{u'v}{c^2}\right).$$

3. The velocity of an object relative to a frame $\mathcal{S}'$ is $\vec{u}'$. If this frame moves with respect to $\mathcal{S}$ with velocity $\vec{v}$, then the general formula giving the velocity $\vec{u}$ of the object in $\mathcal{S}$ is

$$\vec{u} = \frac{1}{\gamma(v) \left(1 + \frac{\vec{u}' \cdot \vec{v}}{c^2}\right)} \left[\vec{u}' + \gamma(v) \vec{v} + (\gamma(v) - 1) \frac{(\vec{u}' \cdot \vec{v}) \vec{v}}{v^2} \right].$$

Show that, if $\vec{v} = v\hat{e}_x$, this gives the three velocity components we derived in lecture, namely,

$$u_x = \frac{u'_x + v}{1 + \frac{u'_x v}{c^2}},$$

$$u_y = \frac{u'_y}{\gamma(v) \left(1 + \frac{u'_x v}{c^2}\right)},$$

$$u_z = \frac{u'_z}{\gamma(v) \left(1 + \frac{u'_x v}{c^2}\right)}.$$